# Brandon Knieriem

**PRINCIPAL ENGINEER AND SYSTEMS ARCHITECT**

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Eight years building and owning production systems, the last nearly six as the sole engineer and technical lead on a student records platform used by every undergraduate college at a 39,414 student university. I design architecture and then build it, from schema through domain layer to interface, and I set the standards and tooling the rest of the team works within. I am usually the person who writes the thing nobody knows how to write yet.

## MILESTONES

- Own the student records platform where a wrong answer decides whether a student graduates; replaced a decade of legacy audit logic.
- Built a design system that is accessible by construction; the in-house WCAG audit last ran clean on all 167 pages.
- Measured my team's delivery with Lorenz curves and Gini coefficients, then fixed it: estimation coverage from 24 to 100 percent.
- Stood up test infrastructure from zero: unit, browser-driven end-to-end, and visual regression in one documented harness.
- First author on a peer-reviewed Springer paper.

## EXPERIENCE

### University of California, Davis

**Technical Lead and Architect, Student Advising Platform** *(Applications Programmer 3)* January 2021 to present

Sole full-time engineer on an eight-repository platform handling degree audit, degree certification, academic petitions, student records, and advising workflows. 814 active staff and faculty users across 25 permission roles, from advisors to deans, plus the full undergraduate student body. Degree certification decides whether a student graduates, so the correctness bar is absolute.

**Built the domain layer that replaced a decade of legacy audit logic.** Two-layer C# architecture separating persistence from domain reasoning, with a decision state model, requirement-scoped authorization, an event-driven notification service, and an append-only SQL decision history.

**Designed the integration architecture across five institutional systems.** Degree audit engine, the Ellucian Banner SIS, institutional messaging, incident management, and work tracking. Covered authentication and access scoping, data governance exposure for student PII, failure handling, and loop prevention on bidirectional sync. Two of the five are third-party systems with no cooperation available.

**Created the design system and led the site-wide redesign onto it.** Modular Sass on BEMIT, a control library that emits WCAG 2.2 AA conformant markup by construction, and a Blazor component library that renders the same system for .NET MAUI clients. The redesign put one shell on about 170 pages, removed Bootstrap and jQuery entirely, applied the UC Davis Brand Communications Guide, and added a token-based dark mode. Wrote the in-house WCAG audit that crawls every page for standards and 320px reflow; the last run was clean on all 167 pages. Live on alpha.

**Prototyped OASIS-API, a faceless REST service on ASP.NET Core.** Per-client hashed keys with scoped roles, Entra ID client credentials planned for production, every request audited. First consumers: advising metrics and a .NET MAUI appointment check-in kiosk on the Blazor library.

**Stood up test infrastructure from zero.** Unit, browser-driven end-to-end, and visual regression in one documented harness, with a 58 area coverage audit.

**Shipped four product areas in 40 sprints.** 1,343 tracked items at 97 percent completion; I held 56.1 percent of all estimated points.

**Diagnosed why delivery was costing what it cost, then fixed it.** Analyzed the full delivery record and quantified workload concentration with Lorenz curves and Gini coefficients, alongside estimation coverage, defect share, and cadence variance. Presented the findings and corrections to executive leadership. The delivery structure I designed moved estimation coverage from 24 percent to 100 percent and concentration from 0.64 to 0.25.

**Technical direction for 14 contributors** including contract engineers, engineers loaned from other departments, and student developers. I facilitate the stand-ups, sprint planning, sprint poker, and retrospectives. Escalation point for anything the team could not resolve.

`C#` `.NET Framework` `ASP.NET Core` `ASP.NET WebForms` `Blazor` `.NET MAUI` `SQL Server` `Ellucian Banner` `JavaScript ES6` `Sass` `NUnit` `Selenium` `Playwright` `Azure DevOps` `PowerShell` `Node.js` `IIS` `ServiceNow` `Figma` `Agile` `Scrum`

### EAM Software Consulting

**Senior Software Engineer** | October 2020 to January 2021

Front end lead for the California Secretary of State website. Rebuilt the voter registration flow with state staff for comprehension and completion. Audited and fixed several hundred accessibility and usability defects across screen reader behavior and navigation.

`Concrete5` `JavaScript` `jQuery` `CSS3` `HTML5` `Atlassian JIRA` `Adobe XD` `JAWS screen reader` `Narrator screen reader`

### AthleteReg (acquired by Outside)

**Full Stack Engineer, UX Researcher, UI Designer** | June 2018 to March 2020

Two-year mobile-first redesign of athletereg.com and its sport portals, covering director and participant features end to end. Built the company's first style guide and component library and established its design process. Wrote 70+ automated visual regression and browser tests. Mentored a junior engineer.

`VB.NET` `Entity Framework` `SQL Server` `JavaScript` `jQuery` `jQuery UI` `CSS3` `Sass` `HTML5` `Telerik` `Gulp` `Backstop.js` `Selenium` `NUnit` `Atlassian JIRA` `Adobe XD` `BrowserStack`

### Acuity Brands Lighting

**Software Engineer Intern** | October 2017 to January 2018

Sustained the SensorView commercial lighting control portal. Built a WiX Toolset installer and integrated it into the package process using MSBuild.

`C#` `ASP.NET WebForms` `SQL Server` `Python` `JavaScript` `CSS3` `MSBuild` `WiX` `Azure DevOps`

### University of New Haven

**CS Teaching Assistant** | September 2016 to September 2017
**CS Research Assistant, UNHcFREG** | June 2015 to September 2017

Led the Artifact Genome Project, an NSA-funded forensic artifact database, analyzing 500+ artifacts. Supervised undergraduate researchers. Built an educational security game used in an NSA-funded cyber camp.

`C` `C++` `Java` `Python` `C#` `JavaScript ES6` `Ruby` `Kali Linux` `Virtual Reality` `Game Development`

### ASSA ABLOY Door Group

**PLC Software Engineer Intern** | September 2016 to May 2017

Assembly line control system projected to eliminate $500,000 in annual scrap loss, programmed in function block diagrams and structured text from stakeholder requirements.

`PLC` `Function Block Diagrams` `Structured Text` `Java` `Raspberry Pi`

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## SELECTED PROJECTS

**PRISM** | A universal image format in dependency-free Rust. Lossless raster with a format-defined reconstruction function plus a binary vector payload. Compresses to 34.0 percent of raw against QOI at 38.2 and PNG at 35.9, with integer-only reconstruction so every viewer scales identically. `Rust`

**Foundering** | Survival-horror roguelite in alpha on Windows, macOS, and Linux. Communicating-vessels flood simulation, a reroutable shipwide power network, seeded always-solvable level generation, and an adaptive narrator, across 453 modeled rooms. `Godot 4` `C#`

**TAN** | Real-time cross-platform audio normalizer. Works blind, with no loudness metadata and no licensed decoder, in a dependency-free Cargo workspace with a client-side browser build that processes live tab audio. MIT licensed. `Rust` `WebAssembly`

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## PUBLICATION

**Knieriem, B.**, Zhang, X., Levine, P., Breitinger, F., and Baggili, I. "An Overview of the Usage of Default Passwords." *Digital Forensics and Cyber Crime* (ICDF2C 2017), LNICST vol. 216, pp. 195-203. Springer, 2018. First author.

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## EDUCATION

**MS Computer Science, Digital Forensics and Security**, University of New Haven, 2018
**MS Criminal Justice, Digital Forensics and Security**, University of New Haven, 2016
**BA Sociology, Criminology**, Eastern Connecticut State University, 2013

**CPUX-F**, Certified Professional for Usability and UX, 2019
**Lean Six Sigma Green Belt**
**People Management Certificate**, UC Davis, ~200 hours